Summary and Recommendations - Customer Churn Exploratory Data Analysis

This exploratory data analysis was conducted on a telecommunications customer dataset containing 7,043 customers and 21 variables related to demographics, services used, billing information, and churn behavior. Multiple visualizations including count plots, stacked bar charts, and percentage-based comparisons were created to identify patterns and key drivers influencing customer churn.

Executive Summary

1. Overall Customer Churn

- The dataset contains 7,043 customers with 21 variables describing demographic, service, and billing attributes.
- The churn distribution indicates:
  - 26.54% of customers have churned
  - 73.46% of customers remain with the service
- This means approximately 1 in every 4 customers leaves the service, indicating a significant retention issue.

2. Gender Distribution

- Female customers: ~49.5%
- Male customers: ~50.5%

Churn patterns across gender are nearly identical, indicating gender has minimal influence on churn behavior.

3. Senior Citizen Analysis

- Only 16.21% of customers are senior citizens, while 83.79% are non-senior customers.
- However, visualizations show senior citizens have a noticeably higher churn proportion compared to non-senior customers, indicating age group may influence service retention

4. Customer Tenure Insights

- The average tenure is 32 months, but churn is concentrated among new customers.
- Customers with tenure below 12 months show the highest churn percentages.
- As tenure increases beyond 24–36 months, the churn proportion drops significantly.

Visualization insight:

  - Early lifecycle customers are the most vulnerable to churn.
  - Long-term customers show much stronger retention behavior.

5. Contract type

It is one of the strongest churn indicators.

Analysis shows:

- Month-to-Month Contracts
  - Represent the largest share of churned customers
  - Approximately 40–45% of these customers churn
- One-Year Contracts
  - Churn drops significantly to roughly 10–12%
- Two-Year Contracts
  - Extremely low churn, approximately 3–5%

Visualization Insight

Customers with short-term flexible contracts are far more likely to leave, while long-term contractual commitment strongly improves retention.

6. Internet Service

The stacked bar charts comparing internet service types reveal:

- Fiber Optic Users
  - Highest churn rate
  - Approximately 40–42% churn
- DSL Users
  - Lower churn levels
  - Approximately 18–20% churn
- Customers without Internet Service
  - Lowest churn proportion

Visualization Insights

Higher churn among fiber users may indicate:

- Higher pricing
- Service expectation gaps
- Customer experience issues

7. Value-Added Service Insights

Charts comparing online security, online backup, device protection, and tech support reveal strong churn differences.

Customers without these services show significantly higher churn percentages.

Examples observed in the visualizations:

- Customers without Online Security
  - Around 40%+ churn
- Customers with Online Security
  - Around 15–20% churn
- Similar trends are observed for:
  - Tech Support
  - Online Backup
  - Device Protection

Visualization Insight

Customers subscribing to support and protection services demonstrate stronger retention, indicating these services increase perceived service value.

8. Streaming and Entertainment Services

- Services like:
  - Streaming TV
  - Streaming Movies

Show moderate differences in churn percentages.

Visualization Insight:

These services do not influence churn as strongly as security or support services.

9. Payment Method Insights

Payment method analysis highlights strong behavioral patterns.

Customers using Electronic Check show the highest churn proportion:

- Electronic Check
  - Approximately 40–45% churn
- Automatic Bank Transfer / Credit Card
  - Around 15–20% churn

Visualization Insight

Customers using automated payment systems tend to remain longer, suggesting billing convenience influences retention.

10. Key Drivers of Customer Churn (Based on Visual Analysis)

The visualizations collectively indicate that churn is primarily influenced by:

- Short customer tenure
- Month-to-month contracts
- Lack of Online Security and Tech Support services
- Fiber Optic internet usage
- Electronic check payment method

Strategic Recommendations

Based on the analysis, churn reduction strategies should focus on:

- Improving early customer onboarding and engagement
- Encouraging long-term contract adoption
- Bundling value-added services such as security and technical support
- Promoting automated payment methods
- Investigating fiber optic service experience and pricing